Physical AI Learning Manual — Lesson 06

Robot Links, Joints, and Degrees of Freedom

Module: 2 — Robot Kinematics

Status: Lesson concepts passed; hands-on Franka joint inspection user-confirmed; USD saved locally, not uploaded or independently inspected.

Simulator: NVIDIA Isaac Sim 6.1 (Windows)

Learning goals

• 1. Distinguish rigid links from the joints connecting them.

• 2. Recognize revolute (rotating) and prismatic (sliding) joints.

• 3. Explain degrees of freedom (DOF) and represent a joint configuration.

• 4. Inspect a real revolute joint's connected bodies, local axis, and limits in Isaac Sim.

Concept 1 — Links and joints

A **link** is a rigid component of a robot (such as a robot forearm). A **joint** connects bodies and specifies allowable relative motion. A hinge-like joint is a simple example. In Lesson 6 the learner correctly classified a rigid metal segment as a link and its connecting rotating hinge as a joint.

Concept 2 — Two joint types

• **Revolute joint:** permits rotation about a specified joint axis. Its coordinate is an angle (theta), expressed in degrees or radians.

• **Prismatic joint:** permits translation along a specified joint axis. Its coordinate is a displacement (d), generally expressed in length units.

A door hinge and a sliding drawer are useful analogies. Both idealized joint types provide one DOF each.

Concept 3 — Degrees of freedom and joint space

A degree of freedom is an independent variable used to describe a mechanism's configuration. For a robot with two independent revolute joints and one independent prismatic joint:

q = (theta_1, theta_2, d_3) DOF = 3

The learner correctly answered both 3 DOF and the mixed joint-variable representation. This count assumes each joint contributes one independent motion variable without additional coupling constraints.

Joint-space coordinates q describe joint configuration, whereas Cartesian coordinates p=(x,y,z) describe a point's location. **Forward kinematics** converts q into positions/poses of links or the gripper; this is a future topic, not yet demonstrated.

Concept 4 — Isaac Sim 6.1: Inspecting a Franka robot

The learner reported loading a Franka asset from the Content Browser and expanding its Stage hierarchy. Observed entries included `panda_link0` through `panda_link7`, `panda_hand`, and finger components. Under `panda_link1` the learner found `geometry` and `panda_joint2`.

The learner inspected `panda_joint2` and reported:

Key distinction: The joint's local attachment orientations (Local Rotation 0 and 1) are not additional degrees of freedom. Axis X designates rotation around the **joint-local** X-axis, which need not align with the world's X-axis.

Motion limits

For a revolute joint with finite angular limits:

theta_min <= theta <= theta_max

angular_range = theta_max - theta_min

With the user-reported values, the nominal configured range is:

101.001 - (-101.001) = 202.002 degrees

This is an arithmetic interpretation of the UI readings, **not an independent validation** of the source asset's authoritative limits or the physical behavior. The user briefly suspected misreading/editing the sign, then reconfirmed the pair. No joint movement or simulation-based limit test was performed.

Practical workflow (reproducible)

• 1. Launch Isaac Sim 6.1 and create a new stage.

• 2. In Content Browser, find and load an articulated Franka robot.

• 3. Expand the Stage hierarchy and locate `panda_link1/panda_joint2`.

• 4. Select `panda_joint2`; inspect joint type, Body 0, Body 1.

• 5. Locate **Revolute Joint** properties and record its Axis and lower/upper limits.

• 6. Do not alter limits or press Play for this observational exercise.

• 7. Save As `lesson_06_robot_joints.usd`.

Actual output: The learner confirmed saving `lesson_06_robot_joints.usd` locally. The USD file was not uploaded and is not inside this documentation package.

Common pitfalls

• The Stage hierarchy is not the same thing as the graph of mechanical joint connections; inspect Body 0/Body 1 relationships.

• Local Rotation 0/1 are attachment-frame orientations, not multiple permitted joint axes.

• Do not assume the joint-local axis points along the corresponding world axis.

• A joint count equals DOF only under the stated independence and constraint assumptions.

• Recorded property values are not proof that the joint physically moved through the full limit range.

Review questions

• 1. How is a link different from a joint?

• 2. Which variable describes a revolute joint? Which describes a prismatic joint?

• 3. What does q=(theta_1,theta_2,d_3) represent?

• 4. What do Body 0 and Body 1 tell you about a physics joint?

• 5. Why does Axis X not guarantee rotation about the world's X-axis?

• 6. Why is reading a joint limit different from testing its actual dynamic range?

Next lesson

Lesson 7 is **not started**. No claims of forward-kinematics competence are made on the basis of this lesson alone.